

数学物理方程(理科版)

郑神州

北京交通大学数学系

Email: shzhzheng@bjtu.edu.cn

July 12, 2020

第二章 波动方程

§3. 初值问题（高维情形）

§ 3.1 解的表达式

一、三维情形(Kirchhoff公式)

(1) 问题: 求解

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} - a^2 \left(\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} + \frac{\partial^2 u}{\partial x_3^2} \right) = f(x, t), & x \in \mathbb{R}^3, t > 0 \\ u(x, 0) = \varphi(x), & x \in \mathbb{R}^3, \\ u_t(x, 0) = \psi(x), & x \in \mathbb{R}^3. \end{cases} \quad (1)$$

(2) 解法: 由简化公式, 只须讨论 $f(x, t) = 0$ 情形. 利用球平均值函数工具, 应用半无界问题解公式, 最终可得解公式.

1. 齐次三维波动方程Cauchy问题

举例: 无限 $\mathbb{R}^3$ 空间中齐次三维波动方程Cauchy问题

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = a^2 \left(\frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} + \frac{\partial^2 u}{\partial x_3^2} \right), & x \in \mathbb{R}^3, t > 0 \\ u(x, 0) = \varphi(x), & x \in \mathbb{R}^3, \\ u_t(x, 0) = \psi(x), & x \in \mathbb{R}^3. \end{cases} \quad (2)$$

考虑: 是否仍然可以用行波法求解?

化为球坐标变换

作球坐标变换

$$\begin{cases} x_1 = r \sin \theta \cos \varphi \\ x_2 = r \sin \theta \sin \varphi \\ x_3 = r \cos \theta \end{cases}$$

或者

$$\begin{cases} r = \sqrt{x_1^2 + x_2^2 + x_3^2} \\ \varphi = \arctan \frac{x_2}{x_1} \\ \theta = \arccos \frac{x_3}{r} = \arccos \frac{x_3}{\sqrt{x_1^2 + x_2^2 + x_3^2}} \end{cases}$$

球坐标变换

一阶偏导数为

$$\begin{aligned}\frac{\partial r}{\partial x_1} &= \frac{x_1}{r}, \quad \frac{\partial r}{\partial x_2} = \frac{x_2}{r}, \quad \frac{\partial r}{\partial x_3} = \frac{x_3}{r}; \quad \frac{\partial \varphi}{\partial x_1} = -\frac{x_2}{x_1^2+x_2^2}, \quad \frac{\partial \varphi}{\partial x_2} = \frac{x_1}{x_1^2+x_2^2}, \quad \frac{\partial \varphi}{\partial x_3} = 0; \\ \frac{\partial \theta}{\partial x_1} &= -\frac{x_1 x_3}{r^2 \sqrt{x_1^2+x_2^2}}, \quad \frac{\partial \theta}{\partial x_2} = -\frac{x_2 x_3}{r^2 \sqrt{x_1^2+x_2^2}}, \quad \frac{\partial \theta}{\partial x_3} = \frac{x_1^2+x_2^2}{r^2 \sqrt{x_1^2+x_2^2}}\end{aligned}$$

将上面结果代入如下的复合函数求导

$$\begin{aligned}\frac{\partial u}{\partial x_1} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial x_1} + \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial x_1} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x_1} \\ \frac{\partial u}{\partial x_2} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial x_2} + \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial x_2} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x_2} \\ \frac{\partial u}{\partial x_3} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial x_3} + \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial x_3} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x_3}\end{aligned}$$

球坐标变换

继续对复合函数的二阶求导 $\frac{\partial^2 u}{\partial x_1^2}, \frac{\partial^2 u}{\partial x_2^2}, \frac{\partial^2 u}{\partial x_3^2}$. 最后得到球坐标系下的三维波动方程:

球坐标系下的三维波动方程

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \varphi^2} = \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2},$$

这里 $u = u(r, \theta, \varphi, t)$.

当解函数 u 不依赖于 θ 和 φ 时, 这时我们称为其为球对称解。

2. 三维波动方程的球对称解

当 u 不依赖于 θ 和 φ 时(即 $u = u(r, t)$), 球坐标系下的三维波动方程可化成

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) = \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2}$$

或

$$\frac{1}{r} \frac{\partial^2 u}{\partial r^2} + 2 \frac{\partial u}{\partial r} = \frac{1}{a^2} \frac{\partial^2 u}{\partial t^2},$$

从而

$$\frac{\partial^2 (ru)}{\partial r^2} = \frac{1}{a^2} \frac{\partial^2 (ru)}{\partial t^2}.$$

这是一个关于 ru 的一维波动方程, 通解为

$$ru = f_1(r + at) + f_2(r - at) \Rightarrow u(r, t) = \frac{f_1(r + at) + f_2(r - at)}{r}.$$

同学们，再见！